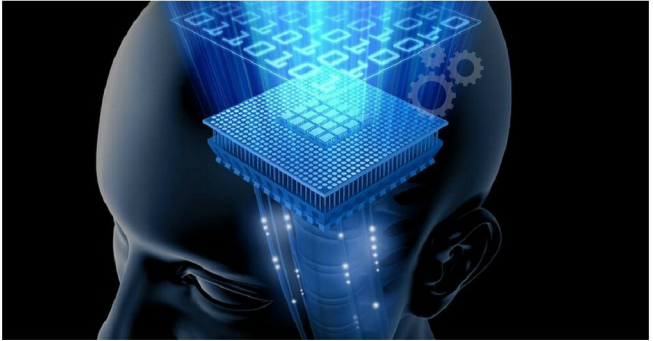


Applying Machine Learning to Real Business Problems

This article is intended to help developers who are working on machine learning initiatives or are planning to set out on this path. It is important to know about some basic concepts and steps when it comes to applying machine learning to real business problems.



Machine learning is a field of artificial intelligence (AI), in which systems are programmed to gather insights from historic data and generate outcomes that can be used for decision making. Machine learning problems can be approached in multiple ways, one of which is collecting lots of data and training the model. We can also derive features from the data and use them to train the model, or pre-process data and transform it into specific formats before using it to train a model. It is not easy to tell which approach will work and give the desired results. Hence, experts suggest some guidelines to approaching a machine learning problem:

- Start with a simple algorithm, then implement and test it with a small data set.
- Plot learning curves to see how it works, and if required, use more data or features.
- Perform error analysis to analyse errors and try to identify a pattern that can help in improving the algorithm.

It is not always recommended that you start with programming a machine learning algorithm but, instead, try it using open source tools such as R or Octave to play around and select the right model that works for particular

types of data.

The objective of this article is to highlight some aspects that are important when selecting an algorithm, setting parameters and diagnosing performance. Some fundamental terms which are referred to in the article are explained below.

Hypothesis: A function that fits the data in linear or non-linear fashion

$$\hat{y} = h_{\theta}(x) = \theta_0 + \theta_1 x$$

Here, the hypothesis function is the equation line; so data will be fitted in a line. We give $h_{\theta}(x)$ values for parameters θ_0 and θ_1 to get our estimated output $\hat{y}$.

Cost function: This is a function that is used to measure the accuracy of the hypothesis function, i.e., how well it fits the data. The function can be described as follows:

$$J(\theta_0, \theta_1) = (\frac{1}{2m}) \sum (\hat{y}_i - y_i)^2 = (\frac{1}{2m}) \sum (h_{\theta}(x_i) - y_i)^2, \text{ for } i=1 \text{ to } m$$

To simplify this, it is $\frac{1}{2} \bar{x}$ where $\bar{x}$ is the mean of the squares of $h_{\theta}(x_i) - y_i$, or the difference between the predicted value and the actual value.